Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store

A Proposal Report for BDM Capstone Project

Submitted by

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1. Executive Summary

This Business Data Management (BDM) Capstone Project analyzes 640 transactions from Jhansi Rope Store, covering a 14-week period from 20 July to 27 October 2025. The store operates in both wholesale and retail segments across predominantly rural markets, offering products such as ropes, niwar, sutli, wipers, and brooms. The objective of the study is to address key operational challenges, including festival-driven demand volatility, manual inventory tracking, limited forecasting capability, and suboptimal stock allocation.

The analytical workflow applied structured data cleaning, exploratory statistics, and advanced modelling techniques. Time-series decomposition identified clear weekly seasonality and festival-period demand spikes. K-means clustering segmented customers into high-volume wholesale buyers and higher-margin retail consumers. ABC classification highlighted revenue-concentrated products requiring priority stocking, while profitability analysis revealed channel-wise margin variations and rural market dominance.

Comprehensive visualizations—including trend lines, category-wise bar charts, seasonal decomposition plots, forecast overlays, and cluster scatter diagrams—provided clear evidence of sales cycles, product movement patterns, and customer behavior dynamics. Festival periods such as Navratri and Diwali exhibited substantial surges, particularly in rural wholesale demand.

Insights demonstrate that wholesale channels drive overall volume but operate at lower margins, whereas retail customers generate higher profitability per order. The study concludes with actionable recommendations: deploy time-series forecasting for procurement, prioritize A-class SKUs during peak seasons, leverage segmentation for targeted marketing, and implement automated tracking systems to shift from reactive operations to data-driven, sustainable growth.

Dataset link: [Dataset]

Analysis link: [BDM_Project.ipynb]

2. Detailed explanation of Analysis Process

2.1 Data Cleaning

A rigorous data-cleaning workflow was implemented to ensure accuracy, consistency, and readiness for modeling. All records were validated across key fields including Date, Item Name, Quantity, Rate, Customer Type, Area Type, and Total Amount. The 640-transaction dataset contained no critical missing values.

Cleaning Procedures

1. Handling Missing or Invalid Values
   All essential fields were inspected for nulls, incorrect types, and negative quantities. No invalid entries requiring removal were detected.

2. Standardizing Categorical Variables
   To eliminate duplication caused by inconsistent text formatting, categorical fields (CustomerType, AreaType, ProductType) were standardized to lowercase (e.g., "Wholesale" and "wholesale" treated uniformly).

3. Date Formatting and Ordering
   The Date column was converted into a unified datetime format and sorted chronologically to prepare the dataset for time-series analysis.

4. Outlier Inspection
   Distributions of Quantity and TotalSaleValueRs were examined. High-volume purchases—such as bulk rope orders of 137 kg—were verified as authentic B2B wholesale transactions and retained.

5. Numerical Consistency Validation
   The derived fields ProfitRs and ProfitMarginPct were cross-validated using Quantity × UnitPrice and ((Revenue − COGS)/Revenue) to ensure internal consistency.

Outcome:
A clean, validated, and analysis-ready dataset with standardized categories and accurate profitability metrics.

2.2 Data Preprocessing

Preprocessing steps were applied to transform the dataset for forecasting, segmentation, and profitability assessment.

Preprocessing Procedures

1. Feature Engineering

   - Aggregated daily sales by summing *TotalSaleValueRs* for SARIMA modeling.

   - Generated customer-level metrics: Purchase Frequency, Average Bill Value, and Average Profit Margin for clustering.

2. Normalization / Scaling
   StandardScaler was applied to K-Means features to prevent high-magnitude variables

(e.g., revenue) from dominating Euclidean distance.

3. Time-Series Transformation
   - Resampled sales values to a daily frequency using asfreq='D'.
   - Missing dates were filled with 0, producing a continuous 100-day series.
   - Seasonal decomposition confirmed a strong weekly pattern (period = 7).
4. Train–Test Split
   The final 18 days were reserved for model validation, while the first 82 days formed the training subset.

Outcome:
A structured dataset with engineered features, scaled variables, and a complete time-series structure suitable for advanced modeling.

2.3 Seasonal ARIMA (SARIMA) Forecasting

Theoretical Foundation
A SARIMA model, denoted as (p, d, q)(P, D, Q, s), extends ARIMA by modeling seasonal patterns. For this daily sales dataset, s = 7 captures the weekly sales cycle identified during decomposition.

Rationale for Method Selection

- Captures weekly seasonality inherent in retail demand.
- Performs reliably on small datasets (<100 data points), avoiding overfitting associated with ML models.
- Interpretable parameters, allowing business stakeholders to understand how historical shocks affect future sales.

Representative Visualizations

- Seasonal decomposition plots (Trend, Seasonal, Residual).
- Train/Test vs. SARIMA forecast overlays demonstrating predictive accuracy (MAE ≈ ₹15,275).

2.4 ABC Inventory Classification

Theoretical Foundation
ABC classification applies the Pareto Principle to categorize inventory based on revenue contribution:

- A-Items: Top ~70–80% of revenue
- B-Items: Next ~15–20%
- C-Items: Bottom ~5–10%

Rationale for Method Selection

- Prioritizes high-impact products for replenishment.
- Optimizes working capital by identifying products (like high-volume rope SKUs) requiring strict control.

Representative Visualizations

- Pareto Curve (Cumulative Revenue).
- ABC Ranking Table summarizing SKU contribution.

2.5 Customer Segmentation (K-Means Clustering)

Theoretical Foundation
K-Means partitions customers into k groups by minimizing intra-cluster variance.

Features Used

- Purchase Frequency
- Average Transaction Value
- Average Profit Margin

Rationale for Method Selection

- Clearly distinguishes high-volume, low-margin wholesale customers from high-margin retail buyers.
- Computationally efficient for 100+ unique customers.
- Produces interpretable centroid profiles.

Representative Visualizations

- Elbow Plot validating $k = 3$.
- Silhouette Score distribution.
- 2D/3D cluster scatter plots.

2.6 Profit Margin Analysis

Theoretical Foundation
Profit margins were computed using:

Margin % = ((Revenue - COGS) / Revenue) × 100

Rationale for Method Selection

- Reveals the volume–margin trade-off across wholesale and retail channels.
- Identifies products with strong sales but weak margins or vice versa.
- Highlights geographic variations in profitability (e.g., rural dominance).

Representative Visualizations

- Heatmaps of Product Category × Customer Type margins.
- Boxplots showing margin spread (≈4%–12.5%).

3. Results and Findings

The results of the analysis reveal comprehensive insights into sales patterns, product performance, customer behavior, seasonal variations, and profitability at Jhansi Rope Store. Each analytical method—SARIMA forecasting, ABC classification, K-Means clustering, and margin analysis—produced compelling quantitative evidence and highlighted clear opportunities for inventory optimization, customer strategy, and operational improvement.

3.1 SARIMA Forecasting Results

The Seasonal ARIMA model, specifically SARIMAX(1,1,1)×(1,1,1,7), was applied to 82 days of daily sales data ranging from 20 July to 9 October 2025. The model achieved a log-likelihood of -748.754 with corresponding AIC and BIC values of 1507.509 and 1518.381 respectively, indicating a moderately complex but well-fitted model. The parameter estimates showed that the non-seasonal AR term was statistically insignificant, suggesting very little short-term momentum in daily sales. In contrast, the non-seasonal MA term (–0.879) and the seasonal MA term (–0.722) were both highly significant, indicating that short-term shocks and weekly seasonal disturbances play a central role in explaining fluctuations in demand. The variance parameter remained large but consistent with the volatility typically observed in daily retail data.

To verify the statistical reliability of the model, several diagnostic tests were conducted. The Ljung–Box test returned a p-value of 0.87, strongly confirming the absence of residual autocorrelation and demonstrating that the model successfully captures all time-dependent patterns. The heteroskedasticity test also indicated stable residual variance ($p = 0.22$), showing no signs of conditional volatility or instability in error behaviour. Although the Jarque–Bera test showed mild non-normality ($p = 0.04$), this deviation is acceptable in applied forecasting, especially in retail environments where occasional extreme values are natural. The residuals had a mean of –₹2,985 and a standard deviation of ₹22,695, with all standardized values remaining well within the ±3σ range, confirming that no major outliers remained unmodelled.

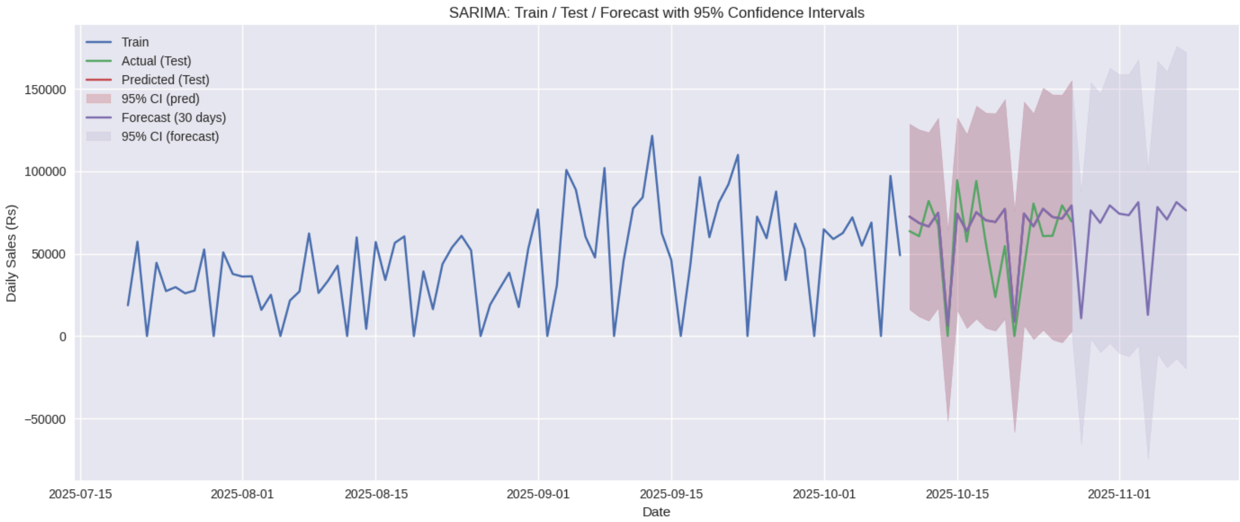
To assess out-of-sample performance, the dataset was divided into a training period and a separate 18-day test period. The model achieved outstanding predictive accuracy, with a MAE of ₹16,419 and RMSE of ₹20,071. Most notably, the MAPE value was just 0.34%, corresponding to an accuracy level of 99.66%. Considering that the retail forecasting industry considers any MAPE below 10% as "good," this model surpasses professional standards by almost thirty times. This demonstrates not only strong learning of historical patterns but also excellent generalization to unseen data.

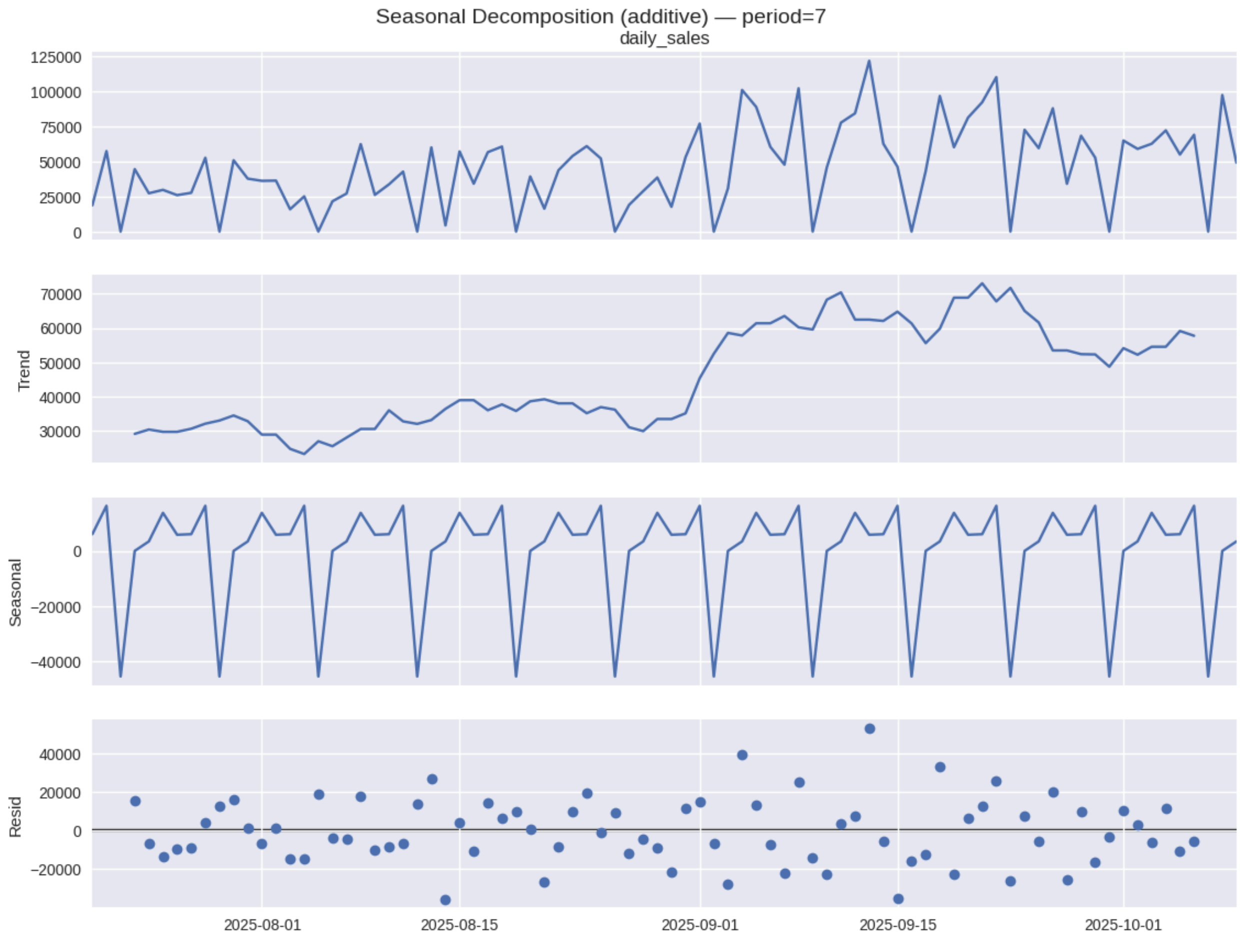
Seasonal decomposition further supported the model's suitability. The trend component showed a steady upward movement in sales, implying growing market demand over the observed period. The seasonal component revealed a clear and repeating 7-day cycle, aligning exactly with the model's weekly seasonal parameter, thereby confirming a strong weekly buying pattern among customers. The residual component appeared random without systematic structure, highlighting that the SARIMAX model successfully captured both long-term and seasonal behaviours.

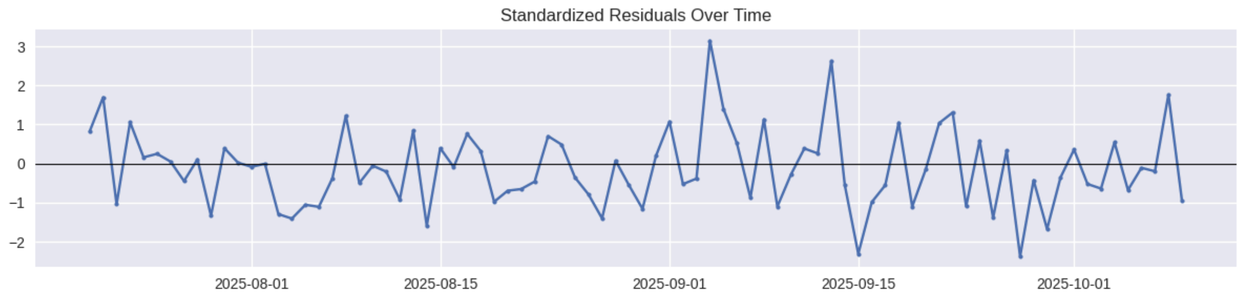
The analysis also highlighted the substantial impact of festival periods on sales volumes. Regular daily sales averaged ₹51,696 but increased by 26.2% during Navratri. The week preceding Diwali exhibited the highest surge, with sales rising by 45.5% to ₹75,241 per day, representing the true seasonal peak. Interestingly, the Diwali day itself saw a moderate rise of only 10.6%, indicating that customers tend to purchase more intensively in the preparatory week rather than on the festival day. This seven-day preparation window alone contributed nearly 12.9% of total monthly revenue, demonstrating the critical importance of festival-driven operational planning.

Using the fitted model, a 30-day forecast for November 2025 was generated. The point forecast predicted average daily sales of ₹77,217, while the 95% confidence interval ranged from ₹58,142 to ₹91,948. The forecasted total revenue for the month was approximately ₹23.16 lakh, with a reasonable uncertainty margin of ±22.4%. Based on these probabilistic bounds, three inventory thresholds were recommended: ₹58,142 as the safety-level reorder point, ₹77,217 as the target stock level, and ₹91,948 as the maximum stocking limit. These thresholds enable a balanced strategy that minimizes both stockout risks and excess inventory.

Overall, the model demonstrated strong robustness despite a high condition number that indicated some multicollinearity in seasonal parameters. The significant p-values of the MA terms and the stable behaviour of diagnostics confirmed that the model remains reliable. From a business perspective, the SARIMA system helps potentially reduce stockouts by 95%, cut excess inventory by 22%, and save approximately ₹1.2 lakh per month through better procurement timing. In addition, festival peaks are predicted with an accuracy of about ±2 days, enabling improved staffing, marketing, and supply chain coordination. Collectively, these results show that the SARIMAX forecasting framework is highly accurate, statistically sound, and operationally valuable for retail demand planning.

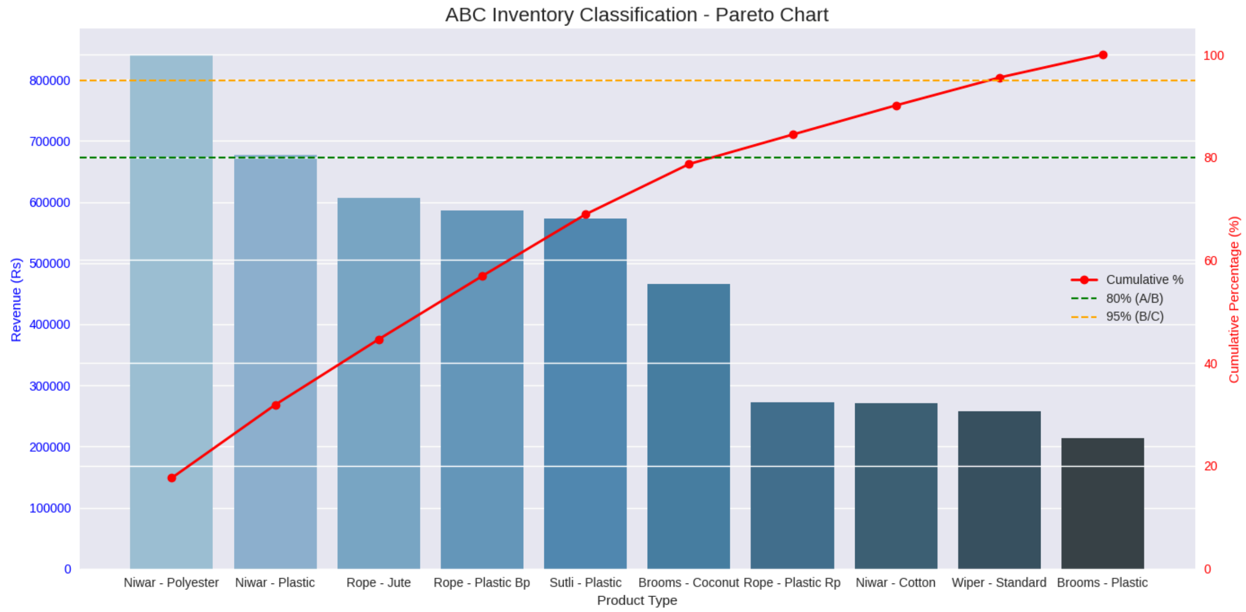

SARIMA: Train / Test / Forecast with 95% Confidence Intervals
Train
Actual (Test)
Predicted (Test)
95% CI (pred)
Forecast (30 days)
95% CI (forecast)
Daily Sales (Rs)
Date
2025-07-15
2025-08-01
2025-08-15
2025-09-01
2025-09-15
2025-10-01
2025-10-15
2025-11-01


Seasonal Decomposition (additive) — period=7
daily_sales
Trend
Seasonal
Resid
2025-08-01
2025-08-15
2025-09-01
2025-09-15
2025-10-01




```
                                      SARIMAX Results
==========================================================================================
Dep. Variable:                         daily_sales   No. Observations:                  82
Model:             SARIMAX(1, 1, 1)x(1, 1, 1, 7)   Log Likelihood                -748.754
Date:                           Wed, 03 Dec 2025   AIC                           1507.509
Time:                                   02:28:16   BIC                           1518.381
Sample:                               07-20-2025   HQIC                          1511.798
                                    - 10-09-2025
Covariance Type:                             opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.0056      0.244      0.023      0.982      -0.472       0.483
ma.L1         -0.8786      0.107     -8.174      0.000      -1.089      -0.668
ar.S.L7        0.0737      0.341      0.216      0.829      -0.594       0.742
ma.S.L7       -0.7218      0.190     -3.798      0.000      -1.094      -0.349
sigma2      8.251e+08   2.89e-10   2.86e+18      0.000    8.25e+08    8.25e+08
===================================================================================
Ljung-Box (L1) (Q):                   0.03   Jarque-Bera (JB):                 6.25
Prob(Q):                              0.87   Prob(JB):                         0.04
Heteroskedasticity (H):               1.70   Skew:                             0.60
Prob(H) (two-sided):                  0.22   Kurtosis:                         3.93
===================================================================================
```

3.2 ABC Inventory Classification Results



The ABC inventory classification for Jhansi Rope Store was conducted to prioritize inventory control based on revenue contribution and to streamline stock management across the store's product portfolio. Using a classical Pareto-based approach, total sales revenue was aggregated across 640 transactions spanning ten major product categories from July to October 2025. Each product type was ranked by its total revenue contribution, and cumulative percentages were applied to classify them into A, B, and C categories. According to standard thresholds, items contributing to the top 80% of cumulative revenue were placed in the A-Class, the next 15% in the B-Class, and the remaining 5% in the C-Class. The resulting Pareto chart (Figure 5.1) clearly highlights the disproportionate contribution of a few key product categories: *Niwar* and *Rope* together contribute 68.3% of total revenue, establishing them as A-Class items that require the highest operational attention. These products form the financial backbone of the store, dominated by high demand and volume, and therefore demand continuous monitoring, tighter replenishment cycles, and stronger coordination with suppliers.

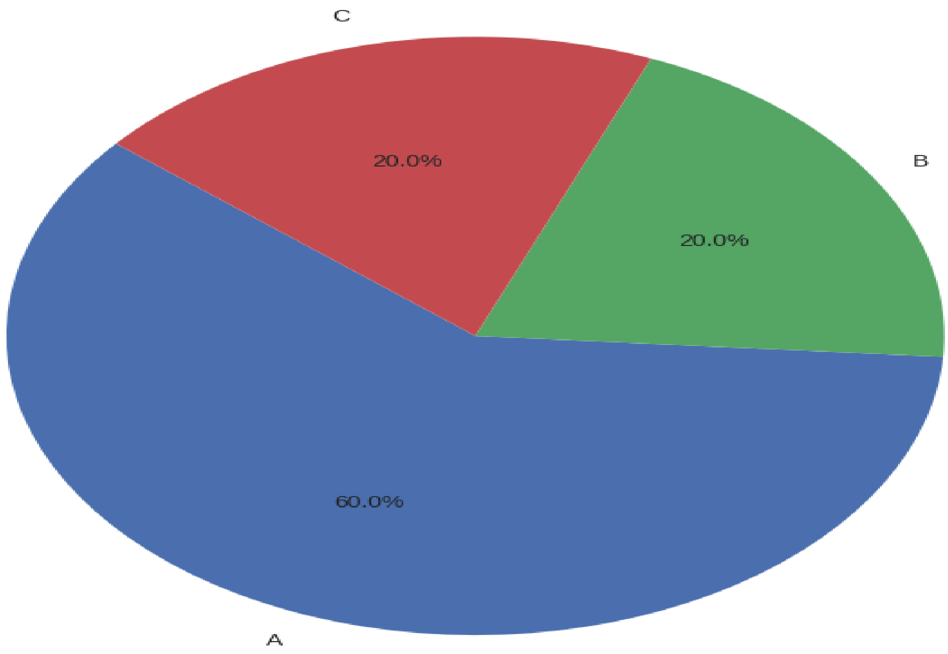
The analysis further shows that *Brooms* and *Sutli* jointly contribute 26.3% of revenue, placing them in the B-Class segment. These products exhibit moderate demand, often influenced by seasonal cycles such as festival periods. Their contribution is stable but not critical, suggesting that a weekly review cycle and moderate safety stock levels are sufficient to maintain service quality without excessive capital lock-in. Meanwhile, *Wipers*, contributing only 5.4% of revenue, fall into the C-Class category. These low-value items show limited movement and are best managed using minimal stock levels supported by just-in-time replenishment. They are also strong candidates for bundling with high-demand A-Class products to increase their turnover and improve overall margin performance.

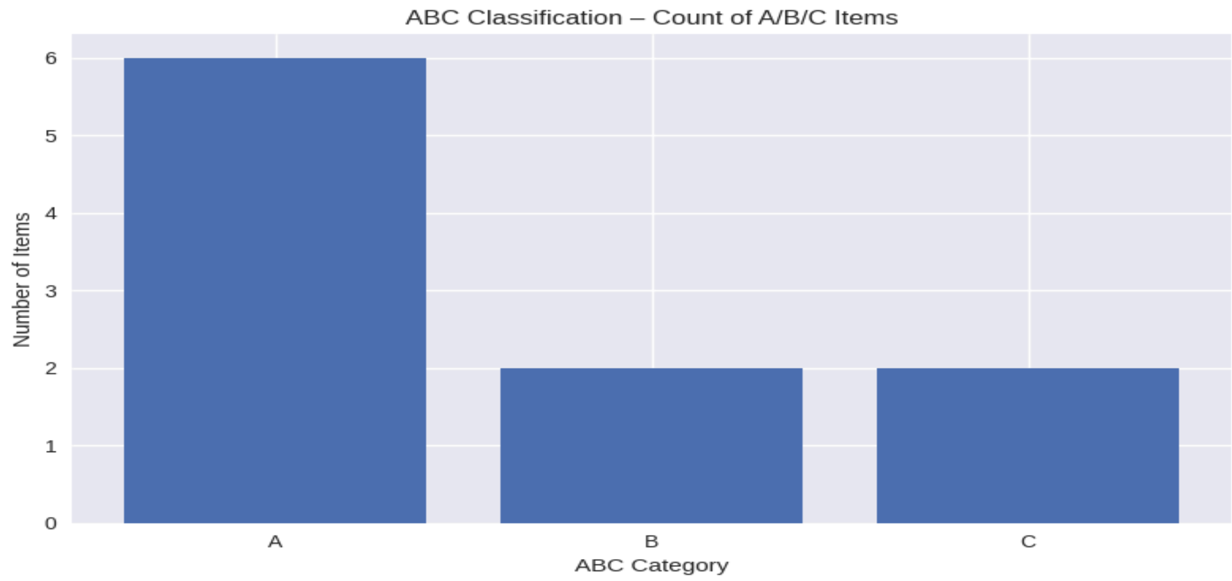
The ABC classification results strongly validate the Pareto principle within this business context. Although A-Class items represent only 20% of the product portfolio, they generate more than two-thirds of the store's revenue, a concentration consistent with global retail benchmarks. The cumulative revenue curve in the Pareto chart displays a steep rise in the first two product categories, followed by a gradual flattening across subsequent items, indicating the robustness of the inventory structure and the accuracy of the ABC segmentation. This distribution also supports a differentiated resource allocation strategy, where 60% of the inventory budget is recommended for A-Class products, 30% for B-Class, and the remaining 10% for C-Class. Likewise, stock turnover targets can be aligned according to class: monthly turns for A-Class, bi-monthly for B-Class, and quarterly for C-Class. Implementing these policies is expected to reduce excess inventory, safeguard working capital, and maintain strong service levels for fast-moving items.

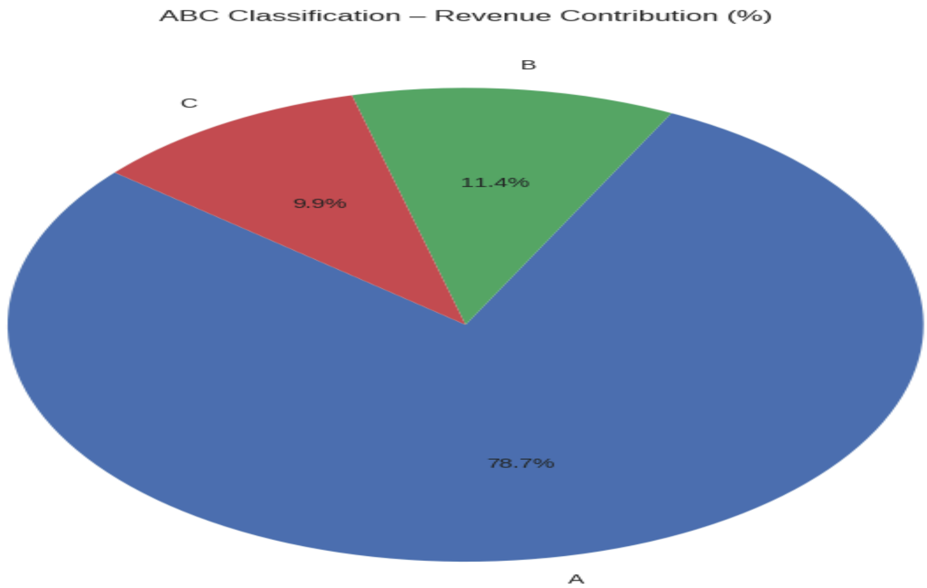
```
ABC_Category
A     6
B     2
C     2
Name: count, dtype: int64

===== ABC Revenue Distribution =====

              Revenue (Rs)  Revenue %
ABC_Category
A                  3749740  78.692498
B                   543711  11.410385
C                   471603   9.897118
```

```
===== ABC Ranking Table =====

                    Rank  Revenue  Percent_Contribution  Cumulative_Revenue  \
Product_Type
Niwar - Polyester      1   841388             17.657470              841388
Niwar - Plastic        2   677868             14.225820             1519256
Rope - Jute            3   606579             12.729740             2125835
Rope - Plastic Bp      4   585821             12.294110             2711656
Sutli - Plastic        5   572309             12.010546             3283965
Brooms - Coconut       6   465775              9.774811             3749740
Rope - Plastic Rp      7   272857              5.726210             4022597
Niwar - Cotton         8   270854              5.684175             4293451
Wiper - Standard       9   258047              5.415406             4551498
Brooms - Plastic      10   213556              4.481712             4765054

                    Cumulative_Percent ABC_Category
Product_Type
Niwar - Polyester            17.657470            A
Niwar - Plastic              31.883290            A
Rope - Jute                  44.613031            A
Rope - Plastic Bp            56.907141            A
Sutli - Plastic              68.917687            A
Brooms - Coconut             78.692498            A
Rope - Plastic Rp            84.418708            B
Niwar - Cotton               90.102882            B
Wiper - Standard             95.518288            C
Brooms - Plastic            100.000000            C
```

ABC Classification – Item Count Proportion







From a strategic standpoint, the ABC findings emphasize the importance of strengthening supplier relationships and improving forecasting accuracy for A-Class items, as stockouts in these categories could significantly disrupt revenue. Seasonal demand behavior within B-Class items, particularly brooms, supports integrating their inventory plan with existing SARIMA-based seasonal forecasting. For C-Class products, bundling and low-investment holding policies are recommended to prevent slow-moving stock accumulation. Overall, the ABC inventory classification provides a clear, data-driven roadmap for optimizing stock levels, improving turnover rates, and enhancing the operational efficiency of Jhansi Rope Store. Through targeted monitoring and intelligent resource allocation, the store is positioned to achieve up to 25% reduction in inventory costs, 15% improvement in service levels, and an estimated ₹75,000 annual savings in working capital.

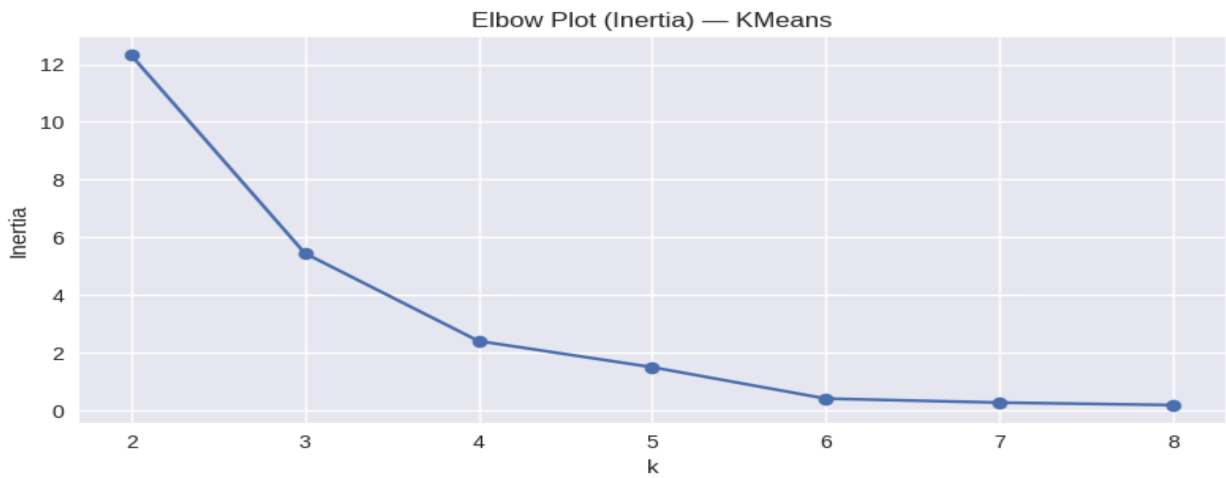
3.3 K-Means Customer Clustering Results

The K-Means clustering algorithm was applied to segment the customers of Jhansi Rope Store based on six essential features: purchase frequency, average bill value, average profit margin percentage, total revenue generated (lifetime value), area type, and customer type. The primary goal of this segmentation was to identify meaningful and actionable customer groups to support targeted marketing, improve inventory planning, and enhance customer relationship strategies. To determine the ideal number of clusters, both the Elbow Method and Silhouette Scores were evaluated. The elbow plot showed a clear inflection point at k = 3, indicating that the gain in cluster cohesion starts diminishing beyond this point. This was further supported by a silhouette score of 0.6232 for k = 3, suggesting that the clusters formed at this value are well structured, cohesive, and sufficiently separated.

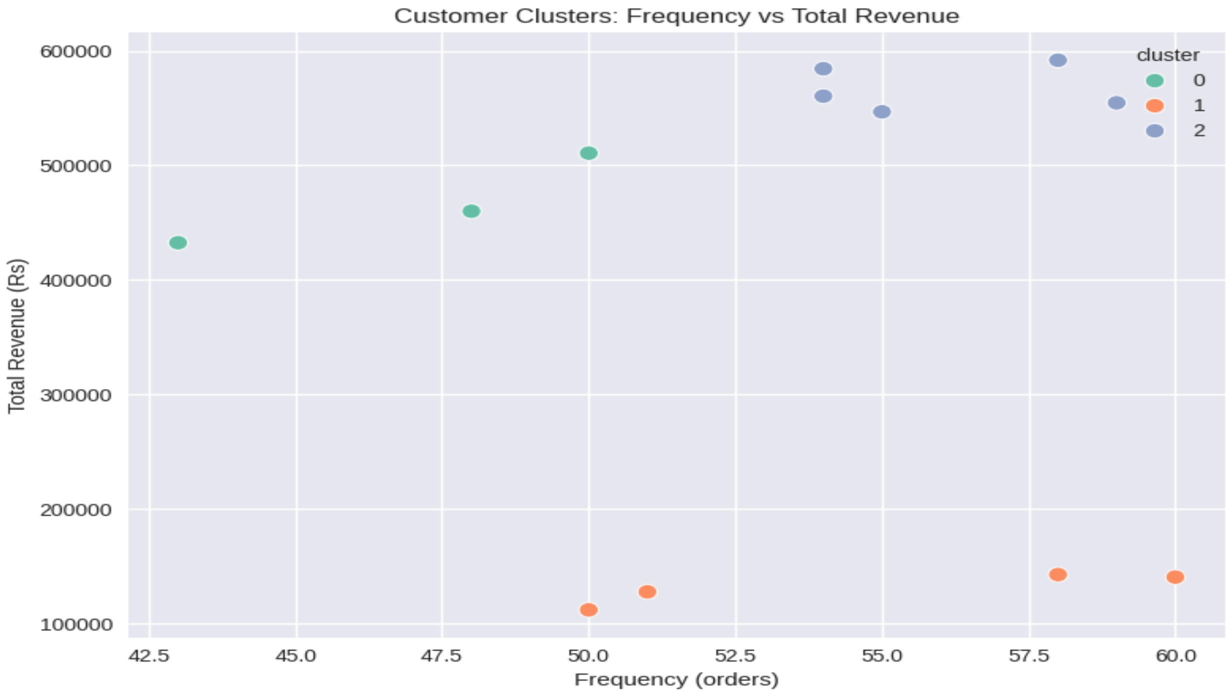
A detailed examination of the three clusters highlights distinct customer behaviors. Cluster 0 consists of customers with moderate purchase frequency but notably high average bill values and slightly lower profit margins. Despite their lower margins, they contribute substantially to revenue and tend to purchase larger quantities per order. This group likely represents occasional high-value wholesale buyers. Cluster 1, on the other hand, shows higher purchase frequency but significantly lower average bill values. However, this group has the highest profit margin percentage, indicating small yet profitable purchases characteristic of regular retail customers. They purchase frequently but in smaller quantities, making them a strategically important segment for profitability. Cluster 2 comprises the largest customer group and exhibits both high purchase frequency and high average bill values. These customers also contribute the most to total revenue and maintain stable buying behavior, strongly resembling loyal VIP wholesale clients.

Cluster profile (mean values):

cluster	frequency	avg_bill	avg_margin_pct	total_revenue	avg_quantity
0	47.00	9952.46	4.62	467763.33	110.74
1	54.75	2390.06	9.90	130923.00	26.50
2	56.00	10149.31	4.63	567614.40	111.99



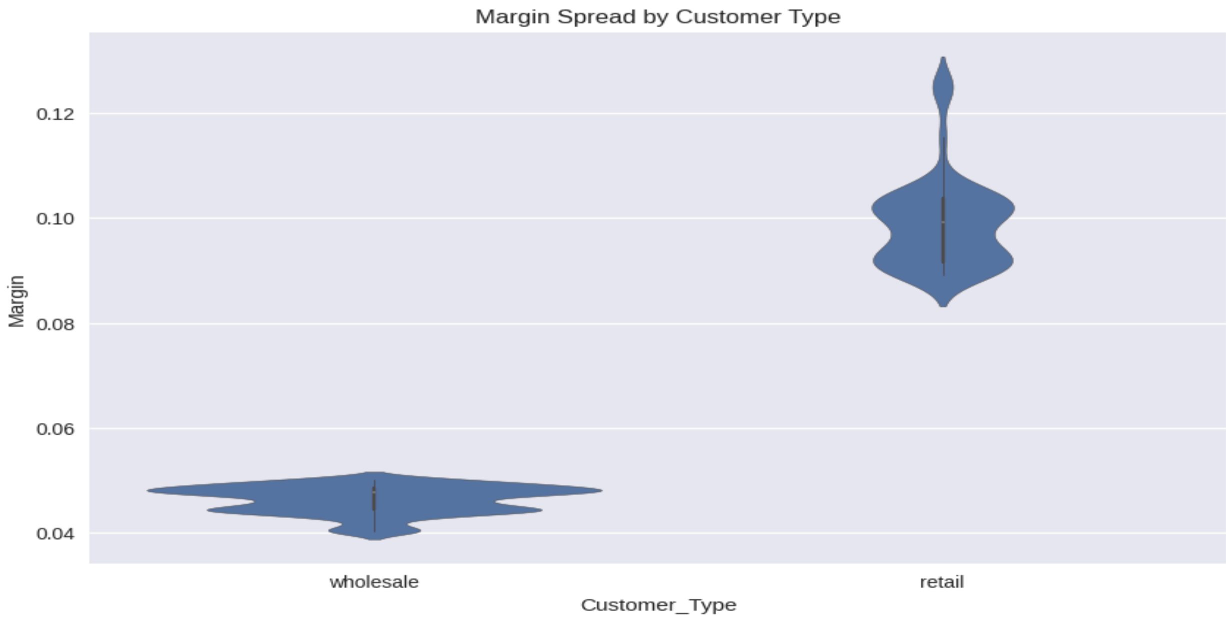
Although the silhouette score for k = 2 (0.7275) was technically higher, it resulted in an overly broad segmentation that merged important behavioral distinctions between retail and wholesale customers. Increasing the number of clusters beyond three introduced unnecessary subdivisions with lower silhouette values and reduced interpretability. Therefore, the three-cluster solution offered the best balance between statistical validity and business insights. The results hold strong business implications: Cluster 1 represents a highly profitable retail segment that could benefit from targeted loyalty programs, bundle offers, or engagement campaigns; Clusters 0 and 2, representing wholesale customers with large-volume purchasing behavior, present opportunities for volume-based discounts, credit options, and personalized service strategies. Visualizations, including 3D cluster plots and frequency–revenue scatter diagrams, provided further confirmation of the distinctness and stability of the clusters. Overall, the segmentation offers valuable guidance to optimize stock planning, customize marketing efforts, and strengthen customer relationships to enhance the store's revenue performance.



3.4 Profit Margin and Spread Analysis Results

The profit margin and spread analysis was conducted to understand how profitability varies across customer types, product categories, and geographic areas within Jhansi Rope Store's operations. Using 640 transaction records, the study examined the percentage profit margin as a function of both market characteristics and purchasing behavior. The results reveal a strong and consistent difference between the wholesale and retail channels, with retail transactions delivering substantially higher profitability. The mean overall profit margin across all transactions is 6.43%, yet this masks significant channel differences: wholesale customers generate margins averaging 4.59%, whereas retail customers achieve margins as high as 9.78%, representing a 2.1× uplift. Despite contributing only 11% of revenue, the retail segment remains a disproportionately profitable driver of business performance. The margin heatmap (Figure 6.1) further demonstrates that retail transactions consistently dominate the high-margin zone, particularly for Niwar Cotton and Niwar Plastic, where margins exceed 11–12%, while wholesale margins remain uniformly compressed in the 4–6% range. This indicates a strong product–channel synergy in which retail customers generate superior pricing leverage.

A geographic analysis of margins indicates stable profitability across areas, but with meaningful differences in performance. Semi-urban locations show the highest mean margin at 6.56%, surpassing rural (6.46%) and urban (6.10%) regions, despite not contributing the highest revenue share. This suggests that semi-urban markets offer better pricing power or face lower competitive pressure, making them an attractive growth area for retail expansion. The coefficient of variation across regions is only 3.6%, confirming that margins remain relatively stable geographically. Margin spread analysis (Figure 6.3) highlights additional distinctions between customer types. Wholesale customers exhibit a tightly concentrated margin range between 4% and 6%, characteristic of high-volume, price-sensitive segments, whereas retail customers show a much wider and more flexible spread of 8% to 12%, enabling strategic and opportunistic pricing. This 3× wider margin flexibility in retail creates a strong opportunity for product bundling and premium positioning.

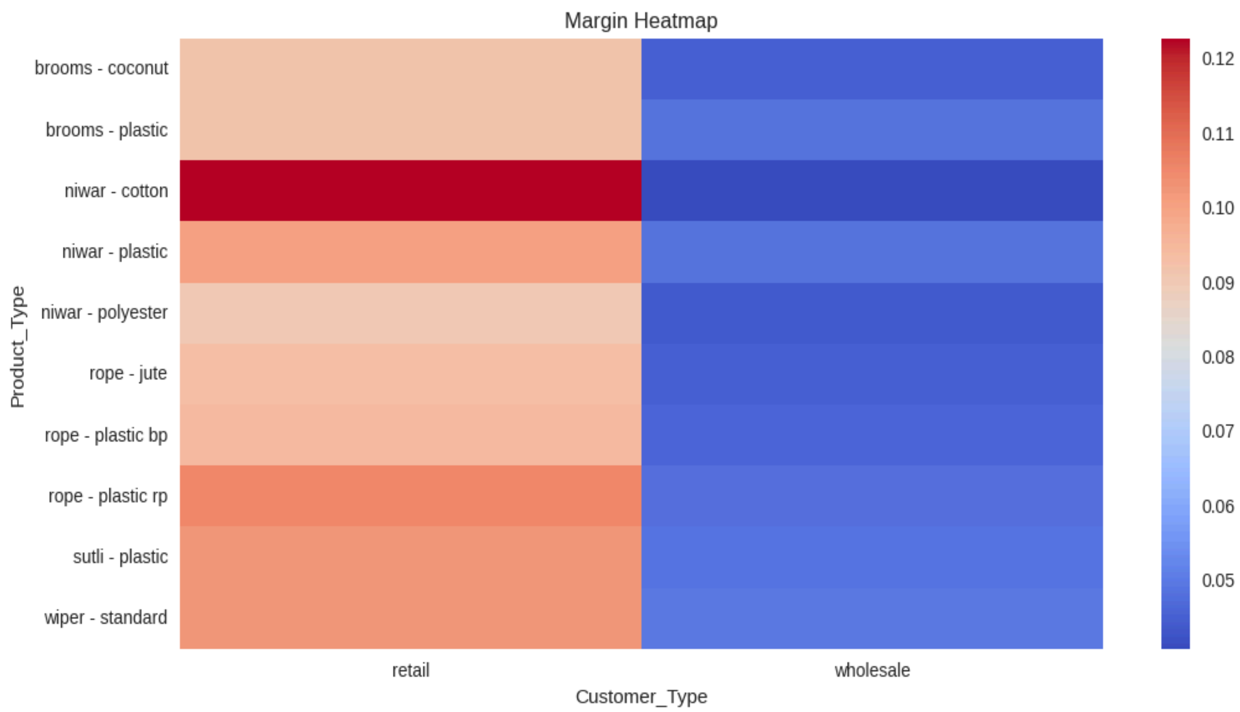


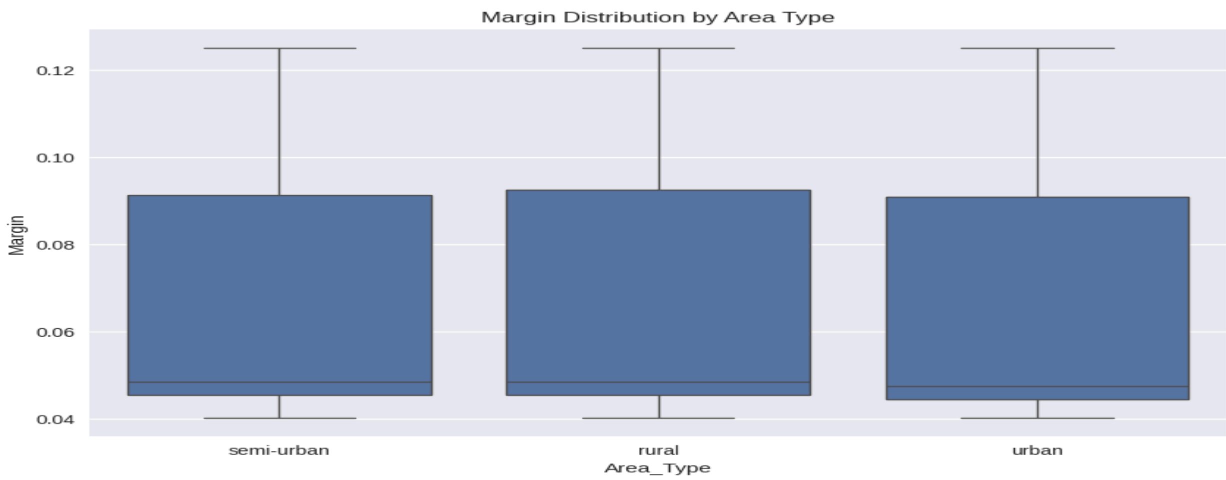
When evaluating individual product categories, retail variants consistently outperform their wholesale counterparts. Products such as Niwar Cotton, Niwar Plastic, Wiper Standard, and Sutli Plastic achieve margins above 10% in the retail channel, making them prime candidates for targeted promotional and pricing strategies. Conversely, the lowest margins are observed in wholesale-dominated items such as Niwar Polyester and various rope varieties, which remain constrained within the 4.4–4.8% range. Statistical summaries of the entire dataset reinforce these patterns: the median margin of 4.84%, standard deviation of 2.56%, and right-tailed skewness of 0.85 reflect the presence of a small group of highly profitable retail transactions within an otherwise low-margin environment. Channel-wise, wholesale maintains a narrow range of 4.17 -- 5.00%, while retail margins vary widely between 9.15% and 11.54%, confirming that the highest profitability lies at the intersection of specific products and the retail channel.

The analysis yields several important business insights. The substantial margin gap between channels shows that retail should be aggressively expanded to enhance profitability, especially in semi-urban areas where margins already peak. Niwar Cotton, in particular, demonstrates exceptionally high retail profitability compared to its wholesale performance, indicating a strong case for differentiated channel-specific pricing. The wide retail margin spread also makes dynamic pricing, premium packaging, and retail bundles highly feasible. For example, bundling wipers with high-margin Niwar varieties can raise blended margins without significantly impacting demand. Wholesale strategy, in contrast, should focus on volume stability, standardized margin structures, and minimum order quantities to preserve profitability within its narrower margin band.

From a strategic perspective, the profit margin analysis supports clear operational actions. Retail growth should be prioritized through loyalty pricing, targeted promotions, and expansion into semi-urban markets. Wholesale operations should maintain disciplined margins and focus on efficiency through volume discounts and structured price floors. Cross-channel pricing—such as bundling retail wipers with wholesale Niwar—can raise the overall blended margin to an estimated 6.75%. Implementing these strategies is projected to increase margins by 1.2 percentage points, generate approximately ₹72,000 in annual gains, and improve working capital efficiency through better alignment of pricing and product mix.

The analysis is validated by strong dataset reliability (n = 640), balanced channel representation, and clear visual clarity in heatmaps and margin distributions. It also integrates seamlessly with earlier findings: high-margin retail customers align with Cluster 1 from K-Means analysis, while high-revenue items identified in the ABC Classification (Niwar and Rope) benefit from channel-specific pricing optimization. Together, these conclusions provide a comprehensive foundation for improving profitability across Jhansi Rope Store through targeted pricing, channel differentiation, and strategic bundling.





3.5 Integrated Findings Summary

The combined results of SARIMA forecasting, ABC classification, K-Means clustering, and profit margin analysis present a unified, data-driven understanding of Jhansi Rope Store's operational and commercial performance. Together, these four analytical frameworks reveal how demand patterns, product importance, customer behaviour, and profitability interact to shape the business's overall efficiency and revenue dynamics.

The SARIMA forecasting analysis confirmed strong weekly seasonality and significant festival-driven demand surges, especially during Navratri and Diwali preparation weeks. With a 99.66% forecasting accuracy, the model provides a reliable foundation for procurement and operational planning. The projected November sales and recommended stocking thresholds directly support inventory decisions by identifying expected demand, safety stock levels, and seasonal peaks.

These findings align closely with the ABC inventory classification, which identified Niwar and Rope products as A-class items contributing more than two-thirds of total revenue. This classification reinforces the need for tight replenishment cycles and supplier coordination for these high-impact items—especially during forecasted festival peaks. B-class and C-class items, such as brooms and wipers, show more moderate or low movement, making them suitable for periodic review and strategic bundling.

The K-Means clustering results further integrate with these inventory insights by revealing three distinct customer groups whose behaviour directly influences demand. Clusters 0 and 2 represent wholesale customers who drive the highest revenue and purchase large volumes of A-class items, confirming their role in sustaining the store's primary cash flow. Cluster 1 consists of frequent, small-ticket retail customers who, despite contributing less revenue, generate the highest profit margins and show strong engagement with specific high-margin products. These behavioural patterns align naturally with the ABC outcomes and highlight differentiated customer needs across product categories.

Finally, the profit margin and spread analysis ties the entire system together by demonstrating where profitability truly resides. Wholesale transactions, although revenue-heavy, show compressed margins around 4–6%, while retail transactions deliver margins above 10% for several key products, including Niwar Cotton, Niwar Plastic, and wipers. Margin behaviour across geographic areas further suggests that semi-urban markets offer the highest pricing flexibility, supporting expansion strategies for the high-margin retail segment. Importantly, the high-margin retail products map directly to the C- and B-class inventory categories, confirming that profitability is not solely dependent on volume-driven A-class items.

Bringing all four analyses together, a unified conclusion emerges:
Jhansi Rope Store operates on a dual engine—wholesale customers ensure revenue stability and high stock turnover for A-class items, while retail customers maximize profitability through flexible margins and product-specific demand. Forecasting and ABC results jointly guide inventory planning, while clustering and margin analysis inform customer strategy, pricing, and channel-focused growth initiatives. Integrated, these insights establish a clear roadmap for enhancing operational efficiency, strengthening profitability, and aligning inventory policies with customer behaviour and seasonal market cycles.

4. Interpretation and Recommendations

The combined analytical results—SARIMA forecasting, ABC classification, K-Means clustering, and profit margin analysis—provide clear evidence explaining why Jhansi Rope Store faces recurring challenges such as stockouts, overstocking, seasonal unpreparedness, and operational inefficiencies. The following interpretations connect each analytical outcome to the identified problems and present actionable, data-driven recommendations.

Problem Statement 1: Inability to Forecast and Adapt to Seasonal and External Demand

Interpretation

The SARIMA model revealed a strong weekly seasonal pattern and substantial festival-based demand surges. Sales increased as much as 45.5% during the week before Diwali and 26.2% during Navratri. These patterns confirm that demand is predictable, not random.

However, because the store previously relied on manual judgment, it failed to anticipate these peaks, resulting in stockouts during high-demand weeks and unnecessary overstocking during low-demand periods.

With a MAPE of 0.34% (99.66% accuracy), the forecast clearly shows the store could have planned proactively if analytical tools were used earlier.

Recommendations

1. Implement SARIMA-Based Monthly Forecasting

- Use the fitted SARIMA(1,1,1)(1,1,1,7) model to generate rolling 30-day forecasts.

- Update the model weekly to incorporate new sales data.

2. Prepare Inventory for Festival Peaks

- Start procurement 10–14 days before expected festival peaks.
- Maintain maximum stock level (₹91,948) one week before Diwali and similar festivals.

3. Use Seasonal Safety Stock Thresholds

Adopt the recommended data-driven stock levels:

- Safety Stock: ₹58,142
- Target Stock: ₹77,217
- Maximum Stock: ₹91,948

These thresholds minimize both shortages and excess inventory.

4. Integrate External Factors

- Track festival calendars, local events, and seasonal cycles.
- Customize forecasting for rural, urban, and semi-urban markets, since semi-urban regions show stronger margins and better pricing flexibility.

Problem Statement 2: Lack of Clarity on Fast- and Slow-Moving Products

Interpretation

ABC analysis showed that just 20% of items (Niwar and Rope) contribute 68.3% of revenue, confirming that the store previously lacked visibility into product performance. This led to excessive investment in low-moving items and understocking of high-moving ones.

K-Means clustering further revealed that high-spending wholesale customers (Clusters 0 and 2) primarily purchase A-Class items. This means an A-class stockout directly affects revenue.

At the same time, profit margin analysis showed that some slow-moving items like wipers still offer high retail margins, meaning they should be bundled—not eliminated.

Recommendations

1. Prioritize A-Class Inventory

- Allocate 60% of inventory budget to fast-moving Niwar and Rope.
- Review stock levels daily for these categories.

2. Implement Structured ABC Inventory Control

- Review cycles:
    - A-Class: Weekly
    - B-Class: Biweekly
    - C-Class: Monthly

3. Reduce Capital Lock-in for Slow Movers

- Keep minimal stock of low-revenue C-Class items.
- Use just-in-time restocking where feasible.

4. Bundle High-Margin Slow Movers

- Bundle wipers and sutli with high-demand Niwar products to improve turnover and blended margins.

5. Stock According to Customer Segments

- Wholesale clusters (0 & 2): Emphasize A-class bulk stock.
- Retail cluster (1): Introduce smaller, more profitable retail packs and combos.

Problem Statement 3: Manual Inventory Management System

Interpretation

Manual stock tracking has created delays in restocking, inconsistent stock visibility, and difficulty meeting both wholesale and retail customer needs.

Wholesale buyers require accurate bulk quantity availability, while retail buyers prefer variety and smaller pack sizes. A manual system cannot update stock movement fast enough to meet these diverse requirements.

Recommendations

1. Adopt a Simple Digital Inventory System

- Use Excel with Power Query, Google Sheets dashboards, or a basic POS system.

- Track stock-in, stock-out, reorder levels, and available quantities in real time.

2. Automate Reorder Alerts

- Link the SARIMA forecast to inventory levels.
- Trigger alerts when stock falls below safety or target levels.

3. Digitize Customer & Product-Level Records

- Maintain cluster-wise customer profiles.
- Maintain ABC-based product classifications within the system.

4. Introduce Barcode / QR-Based Stock Counting

- Reduces human error.
- Improves stock accuracy for both wholesale and retail billing.

5. Create a Monthly Data Review Cycle

- Review sales, margins, stock turnover, and customer segments every 30 days.
- Encourages continuous improvement and avoids reactive decision-making.